API INTEGRATION DOCUMENTATION

CarPark Ed. 7 — Hubtel MoMo Flow

How the USSD app interfaces with Hubtel Direct Receive Money API

Server IP **34.121.254.209** Merchant **2038733** API ID **BrAIO3J**

USSD Code ***899*92** Date **2026-04-22**

END-TO-END PAYMENT FLOW

Customer

1

Customer dials *899*92

Customer dials the USSD short code on MTN Ghana. mNotify routes the session to our server via POST to the callback URL.

```
POST https://api.winbig.bedriften.xyz/api/v1/ussd/callback
msisdn=233244XXXXXX  // customer phone
sequenceID=abc123  // session ID
network=mtn  // network carrier
```

USSD Menu

2

Customer selects ticket type

Customer navigates the menu: Buy Ticket → 1-Day Pass / 2-Day Pass / Extra WinBig Entry → Confirm & Pay.

```
// Menu paths
1*1*1 → 1-Day Pass (GHS 100, 1 draw entry)
1*2*1 → 2-Day Pass (GHS 180, 2 draw entries)
1*3*N*1 → N Extra WinBig entries (GHS 20 each)
```

Our Server

3

Tickets saved to database (status: pending)

Before payment is triggered, tickets are written to PostgreSQL immediately. Access pass (CP-ACC-XXXXXXXX) and draw entries (WB-ENT-XXXXXXXX) are pre-generated and stored with **payment_status = pending**.

```
// Ticket types saved
game_type: ACCESS_PASS → CP-ACC-XXXXXXXX
game_type: DRAW_ENTRY → WB-ENT-XXXXXXXX
```

```
payment_ref: a3f2c1d0e4b5 ...  // UUID4 hex, 32 chars
payment_status: pending
```

Our Server

4

USSD session closed — background thread fires

The USSD session is closed immediately (continueFlag: 1) with a confirmation screen. A background thread waits 3 seconds then calls Hubtel.

```
// USSD response sent to customer
Processing payment ...
Pass: CP-ACC-XXXXXXXX
Draw Entries: WB-ENT-XXXXXXXX
Enter MoMo PIN when prompted.
SMS sent on confirmation.

continueFlag: 1 // closes session
```

Hubtel API

5

POST to Hubtel — initiate MoMo charge ResponseCode 0001

Our server POSTs to Hubtel's Direct Receive Money endpoint using Basic Auth. Amou in GHS float. ClientReference is the UUID stored in the DB.

```
POST
https://rmp.hubtel.com/merchantaccount/merchants/2038733/receive/mobilemon
Authorization: Basic Base64(BrAlO3J:<API_KEY>)

{
  "Amount": 100.00,
  "Title": "CarPark Ed. 7",
  "Description": "Car Park ticket / WinBig draw entry",
  "PrimaryCallbackUrl": "https://api.winbig.bedriften.xyz/payment/webhook"
  "ClientReference": "a3f2c1d0e4b5f6a7b8c9d0e1f2a3b4c5",
  "CustomerMsisdn": "233244XXXXXX",
  "Channel": "mtn-gh"
}

// Hubtel response
{ "ResponseCode": "0001", "Message": "Transaction pending ... ",
  "Data": { "TransactionId": "17f68224 ... " } }
```

MoMo Prompt

6

Customer receives MoMo PIN prompt

MTN Ghana sends a USSD push prompt to the customer's phone asking them to enter their MoMo PIN to approve the payment.

```
// MTN push to customer's handset
"Approve GHS 100.00 payment
```

```
to CarPark Ed. 7?
Enter PIN to confirm."
```

Hubtel

7

Hubtel sends webhook callback to our server

After the customer approves or rejects, Hubtel POSTs to our **PrimaryCallbackUrl** with the final transaction result.

```
POST https://api.winbig.bedriften.xyz/payment/webhook

// Payment approved (ResponseCode 0000)
{
  "ResponseCode": "0000",
  "Data": {
    "ClientReference": "a3f2c1d0e4b5f6a7b8c9d0e1f2a3b4c5",
    "Amount": 100.00,
    "CustomerMsisdn": "233244XXXXXX"
  }
}

// Payment rejected (ResponseCode 2001)
{ "ResponseCode": "2001", "Data": { "ClientReference": "…" } }
```

Database

8

Ticket status updated in database

Our webhook handler looks up tickets by **ClientReference** and updates their payment status.

```
// On ResponseCode 0000
UPDATE tickets SET payment_status='completed'
WHERE payment_ref='a3f2c1d0…'

// On ResponseCode 2001
UPDATE tickets SET payment_status='failed'
WHERE payment_ref='a3f2c1d0…'
```

mNotify

9

Confirmation SMS sent to customer Done

On successful payment, an SMS is sent via mNotify confirming the ticket details.

```
// Day Pass SMS
CarPark payment confirmed!
Pass: CP-ACC-XXXXXXXX
Valid: Day 1
Draw entries: 1
```

```
Draw: 03 May 2026
Good luck!
```

HUBTEL RESPONSE CODE REFERENCE

CODE	HTTP	DESCRIPTION	REQUIRED ACTION
0000	2xx	Transaction processed successfully.	Mark completed, send confirmation SMS.
0001	2xx	Request accepted. Callback will be sent on final state.	Wait for webhook callback. No action needed.
2001	2xx	MTN MoMo user reached counter/balance limits, insufficient funds, or missing permissions. Vodafone Cash failed. Wrong PIN entered. USSD session timeout. Strange characters (&*!%@) in description. Number does not match channel. Account Holder with FRI Not Found.	Mark tickets as failed. Customer may retry by dialling *899*92 again.
4000	4xx	Validation errors. Something is not quite right with the request.	Check all required fields in the payload are present and correctly formatted.
4070	4xx	Unable to complete payment. Fees not set for given conditions.	Ensure minimum amount is passed, or contact Hubtel relationship manager to set up fees.
4101	4xx	Business not fully set up to receive payments. API keys missing "mobilemoney-receive-direct" scope. Client request keys do not match API keys on business.	Contact Hubtel Retail Systems Engineer to enable Receive Money scope. Verify Basic Auth key and POS Sales ID in endpoint.
4103	4xx	Permission denied. Account not allowed to transact on this channel.	Contact Hubtel Retail Systems Engineer.
403	4xx	Server IP not whitelisted on Hubtel.	Request IP whitelist from Hubtel support.

CREDENTIALS & CONFIGURATION

API ID (USERNAME)
`BrAlO3J`

POS SALES ID
`2038733`

CALLBACK URL
`api.winbig.bedriften.xyz/payment/webhook`

CHANNEL (MTN GHANA)
`mtn-gh`

AUTH METHOD
`Basic Auth (Base64)`

AMOUNT FORMAT
`GHS float (e.g. 100.00)`

> ⚠ **CHARGE_GHS_1 = True** is currently active — all charges are sent as GHS 1.00 for testing. Set **CHARGE_GHS_1 = False** in app.py before going live.

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